Third System
AI Presence Index

PM Software × Google Trends

Construct Validity, $t_1 \rightarrow t_2$

May 2026 · Pablo Ulpiano Gonzalez Castro, Third System

Independent measurement for the AI mediation layer.

Does AI Presence correspond to consumer search interest? v0.11 tests this on 18 project management software brands across both, at two longitudinal waves seven days apart.

The correlation is positive, statistically significant, and reproducibly stable across both waves (Spearman $\rho = 0.50$ at t_1, 0.48 at t_2; $p < 0.05$ one-tailed at both; $|\Delta\rho| = 0.02$ — H2 confirmed). It narrowly misses the pre-registered moderate-to-strong threshold ($\rho > 0.5$) by 0.004 at t_1 and weakens further under covariate control (partial $\rho \approx 0.41$ after age + tier control). The construct's divergence from consumer search is concentrated and identifiable: only one of three AI-top-three brands appears in the Trends top-five, at both waves. The leadership zone is where AI applies a tighter category boundary than consumers do.

The AI Availability Score (AIAS) research programme has, through v0.6–v0.10, established a methodologically rigorous LLM-side measurement of brand presence in AI-mediated decision contexts. What it has not yet established — and what the framework's external validity ultimately rests on — is that this LLM-side measurement corresponds to what consumers actually do. v0.11 is the first construct-validity test in the programme: a single-category pilot against an external behavioural validator (Google Trends search interest).

The result is informative in a way that a clean confirmation or a hard rejection would not be. AI Presence and Google Trends search interest are positively correlated (**Spearman $\rho = 0.50$ at t_1, 0.48 at t_2; $p < 0.05$ one-tailed at both waves**) and stably so across the seven-day longitudinal interval (**$|\Delta\rho| = 0.02$**). The two waves measure essentially the same relationship. But the correlation narrowly misses the pre-registered moderate-to-strong threshold ($\rho > 0.5$) at t_1 by 0.004, and weakens further under covariate control (**partial $\rho \approx 0.41$ after age + tier control**).

The leaderboard test (H3) localises where the construct fails. At both waves, only **one of three AI-top-three brands appears in the Trends top-five**. The construct's divergence is concentrated at the top of the AI Presence distribution — exactly the zone where AI mediation carries the most predictive weight for purchase decisions. **Linear** is AI's #1 recommendation in the category (86.5% AI Presence) with negligible consumer search interest (Trends value 1.88, Asana indexed to 100). **Todoist** is the inverse: 1.04% AI Presence with Trends value 28 — comparable to GitHub Projects, which receives 40× more AI Presence. Different brands, same diagnostic: AI Presence is not interchangeable with consumer search.

The substantive interpretation is that AI applies a tighter and partly-different category boundary than consumer search does. Linear, framed by AI as developer-focused PM, falls inside the AI category; Todoist, treated by AI as a personal task manager, falls outside it. Both are real PM tools by ordinary consumer reasoning. The construct of AI Availability is ***correlated but not equivalent*** to Mental Availability — which is the position the Tri-System Brand Growth framework articulates theoretically and which v0.11 now supports empirically in a single category.

Phase 3 expansion (v0.12, three categories) will test whether this divergence pattern is specific to project management software or generalises. Categories where the leadership zone is stable across both AI and consumer search (premium olive oil is a candidate) may yield $\rho > 0.5$. Categories where AI applies idiosyncratic boundaries (personal finance with its phantom-brand effects) may yield ρ further below.

What We Measured

v0.11 measures the cross-sectional correlation between two independent brand-level signals: per-brand AI Presence rate (drawn from v0.9's deposited matched-subset LLM measurements) and per-brand Google Trends search interest (acquired fresh at v0.11 under a pre-registered protocol).

AI Presence input. Per-brand AI Presence rates for 18 PM software brands at two waves: t_1 = v0.6 collection window (29–30 April 2026); t_2 = v0.9 re-baseline window (7 May 2026). Both waves use the matched-model subset (Claude Sonnet 4.6 + GPT-5.4-mini) restricted to PM software per the v0.9 deposit's H2 leaderboard. n = 96 responses per wave.

Google Trends validator. Daily search-interest indices for the same 18 brands acquired in a single locked session on 10 May 2026 via SerpAPI's Google Trends engine. Five pivot bundles (Asana as anchor; topic IDs where available, distinctive strings or '<brand> project management' compound queries where not) across two regions (Worldwide primary; US sensitivity). Pivot-rescaling converts the bundle-local 0–100 indices into a single cross-brand-comparable scale where Asana 100.

Two waves. t_1 Trends window: 27 April – 3 May 2026 (centred on the v0.6 collection window). t_2 Trends window: 4 May – 10 May 2026 (centred on the v0.9 collection window). Windows are fully disjoint at the 3/4 May boundary with identical Monday–Sunday composition.

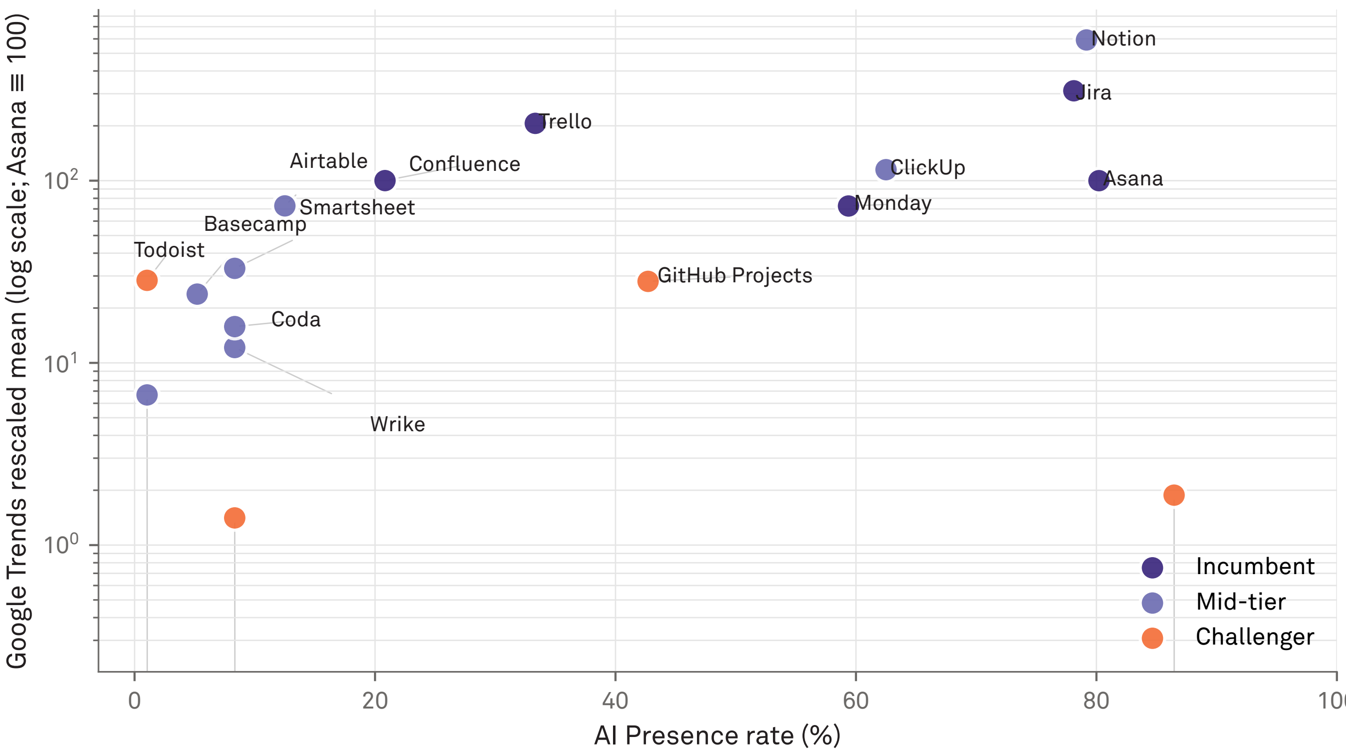
Pre-registration. The protocol was locked at git tag ***v0.11-prereg*** (commit f20ade8) prior to any acquisition call against the wave windows. Four hypotheses with numerical thresholds: H1 cross-sectional construct validity (ρ > 0.5 at both waves, primary); H2 stability (|Δρ| 0.15); H3 leaderboard top-3 top-5 at both waves; H4 covariate-controlled (partial ρ > 0.5 after age + tier control). One brand (Shortcut) was excluded pre-acquisition under rule E1a; one brand (Height) was retained as a phantom brand following the rebrand-period convention from v0.7 / v0.10.

FINDING 01

The construct is positive, significant, and just-misses

H1 — AI Presence × Google Trends, t_1 (Worldwide)

n = 17. Spearman ρ = 0.496 (p_{1t} = 0.0214). Pearson r = 0.544 (sensitivity).



Source: Third System AI Presence Index v0.11 · Google Trends acquisition 2026-05-10 · Pre-reg locked at v0.11-prereg (commit f20ade8).

Figure 1 · Per-brand AI Presence (matched subset, Sonnet 4.6 + GPT-5.4-mini, PM software) against Google Trends rescaled mean at t_1. Worldwide region. Log-y axis. Pivot brand Asana indexed to 100 by construction. Spearman ρ = 0.496 (p_{1t} = 0.021); Pearson r = 0.544. n = 17.

Spearman ρ = 0.496 at t_1 with one-tailed p = 0.021. Positive and significant — but 0.004 below the pre-registered moderate-to-strong threshold.

Per-brand AI Presence rates correlate positively with per-brand Google Trends rescaled means at t_1: Spearman ρ = 0.496 (n = 17 brands clearing the at-acquisition eligibility rule). One-tailed p = 0.021 against the null of zero correlation. The directional claim of the construct — AI Presence and consumer search interest co-vary at the brand level — is statistically supported.

The pre-registered threshold was ρ > 0.5, calibrated against the conventional moderate-to-strong correlation benchmark in behavioural research. v0.11 misses this bar by 0.004. The pre-registration discipline gives this near-miss its weight: a post-hoc adjustment to ρ [illegible] 0.49 would convert the result, but the bar is what it is. **H1 is falsified at the locked threshold.**

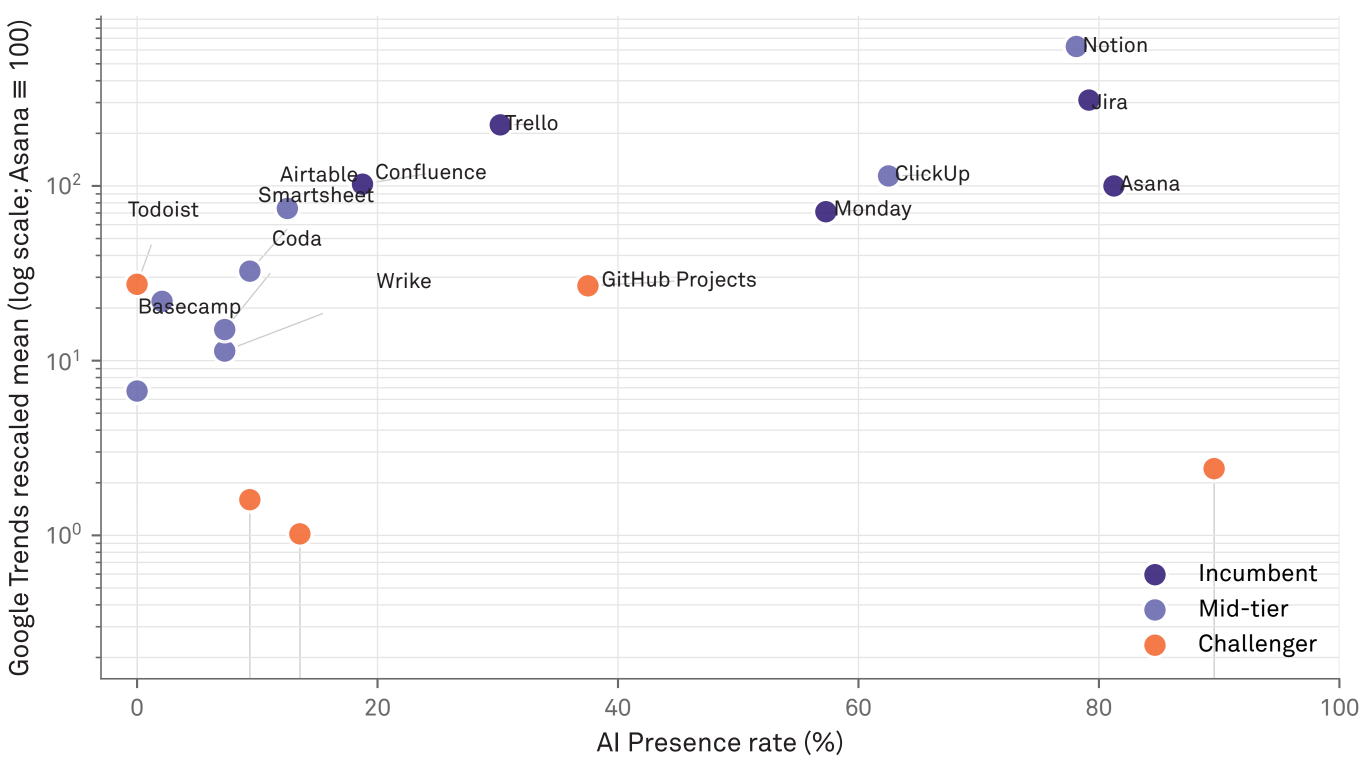
The Pearson sensitivity, in contrast, clears the same bar (r = 0.544 at t_1; r = 0.533 at t_2). This Pearson-vs-Spearman disagreement is not a methodological glitch. Pearson weights large absolute differences; Notion's Trends value of 591.95 (≈ 6× Asana) anchors the linear fit and pulls Pearson up. Spearman treats Notion as a single high rank and is in turn pulled down by the rank-1-AI / rank-low-Trends inversions at the top of the AI distribution (Linear; see Finding 3). The two statistics tell different parts of the same story.

FINDING 02

The just-miss is stable, not noise

H1 — AI Presence × Google Trends, t_2 (Worldwide)

n = 18. Spearman ρ = 0.476 (p_{1t} = 0.0228). Pearson r = 0.533 (sensitivity).



Source: Third System AI Presence Index v0.11 · Google Trends acquisition 2026-05-10 · Pre-reg locked at v0.11-prereg (commit f20ade8).

Figure 2 · Same shape at t_2. The cross-wave visual stability is the H2 finding: Spearman ρ = 0.476; |Δρ| from t_1 = 0.020. Whatever the construct is, it reproduces.

ρ_{t_2} = 0.476 against ρ_{t_1} = 0.496. |Δρ| = 0.020 — far below the 0.15 stability tolerance. H2 confirms: whatever H1 measured, it measured reproducibly.

The H1 correlation magnitude is essentially identical at t_2: Spearman ρ = 0.476, |Δρ| = 0.020 against the t_1 value. The pre-registered tolerance was |Δρ| ≤ 0.15, calibrated to absorb sampling variance at small n while detecting meaningful instability. **H2 is confirmed** with almost an order of magnitude of margin.

The H2 confirmation is the load-bearing positive finding in v0.11. It rules out the interpretation that the H1 near-miss is wave-specific noise: the two waves' correlations sit on top of each other. The Pearson sensitivity tells the same story (r_{t_1} = 0.544; r_{t_2} = 0.533; |Δr| = 0.011).

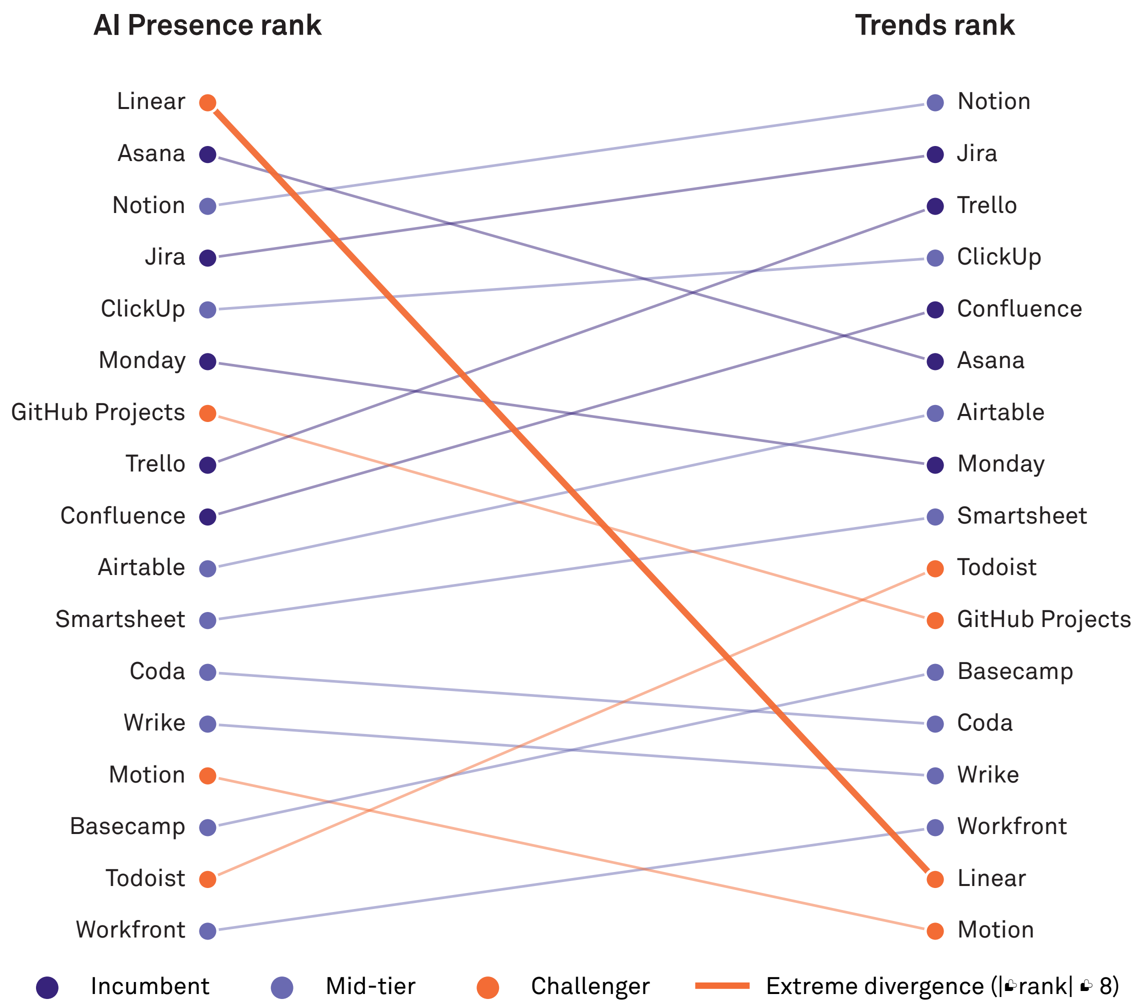
The US-only sensitivity replicates the Worldwide pattern (ρ_{t_1} = 0.489; ρ_{t_2} = 0.455; |Δρ| = 0.034). Nothing region-specific is driving the result, and the within-wave US-vs-Worldwide difference is itself smaller than the H2 tolerance.

FINDING 03

Where the construct diverges: AI's category boundary

H3 — AI Presence rank vs Google Trends rank, t_1 (Worldwide)

Lines connect each brand's rank by AI Presence (left) to its rank by Trends (right). Steep slopes = construct divergenc



Source: Third System AI Presence Index v0.11 · Google Trends acquisition 2026-05-10 · Pre-reg locked at v0.11-prereg (commit f20ade8).

Figure 3 · Slope chart: each brand's rank by AI Presence (left) connected to its rank by Trends (right) at t_1. Steep slopes indicate construct divergence. Extreme cases (|Δrank| ≥ 8) highlighted in copper. Linear (top-left) and Todoist (mid-right) anchor the divergence diagnostic discussed in Finding 3.

1 of 3 AI-top-three brands appear in the Trends top-five — at both waves. Linear (AI rank 1, Trends rank 14) and Todoist (AI rank 17, Trends rank 9) bracket the same diagnostic from opposite ends.

H3 is falsified. At t_1, the AI-top-three is Linear, Asana, Notion. The Trends-top-five is Notion, Jira, Trello, ClickUp, Confluence. Only Notion appears in both. At t_2, the AI-top-three is Linear, Asana, Jira; the Trends-top-five is unchanged from t_1. Only Jira appears in both. The divergence concentrates at the top of the AI Presence distribution — the zone where AI Presence carries the most predictive weight for purchase decisions.

The Linear paradox. Linear has the highest AI Presence in the registry (86.5% at t_1, 89.6% at t_2) and the second-lowest Trends value (rescaled mean 1.88 at t_1; Asana indexed to 100 by construction). AI systems strongly recommend Linear despite the brand's negligible consumer search interest. The bare query 'Linear' returns Google Trends results dominated by linear-algebra textbooks; the brand-disambiguated query ***'linear project management'*** returns the residual signal reported here.

The Todoist inverse. Todoist has the second-lowest AI Presence at t_1 (1.04%; effectively zero at t_2) and Trends value 28.38 — comparable to GitHub Projects (28.01), which receives 42.7% AI Presence. At similar consumer search interest, AI grants GitHub Projects 40× more recommendation share. The mechanism is plausibly category-boundary: AI frames Todoist as a personal task manager rather than a project management tool, and excludes it from PM recommendations.

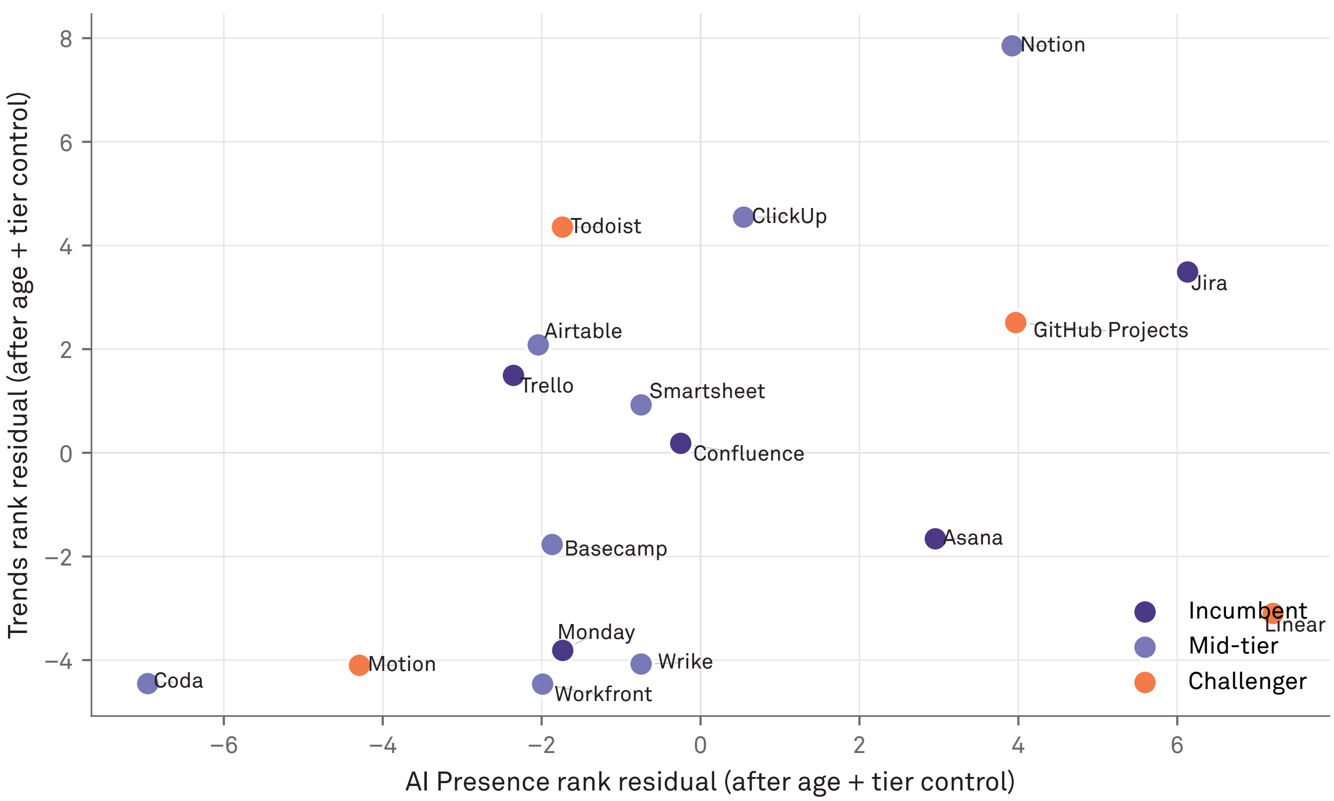
Both diagnostics support the same substantive finding: **AI applies a tighter and partly-different category boundary than consumers do.** Linear is inside AI's PM category but barely inside consumers' search-revealed one; Todoist is outside AI's PM category but well inside consumers' search-revealed one. The construct of AI Availability is empirically distinct from Mental Availability — and the distinction is structural, not noise.

FINDING 04

Brand age and tier absorb part of the H1 signal

H4 — Covariate-controlled construct validity, t_1 (Worldwide)

Rank residuals after OLS on rank-transformed brand age + competitive density.
Partial Spearman ρ = 0.407 (p_{1t} = 0.0663).



Source: Third System AI Presence Index v0.11 · Google Trends acquisition 2026-05-10 · Pre-reg locked at v0.11-prereg (commit f20ade8).

Figure 4 · H4 partial correlation visualisation. Rank residuals on both axes after OLS on rank-transformed brand age and tier ordinal. Partial Spearman ρ = 0.407 (p_{1t} = 0.066), missing both the moderate-to-strong threshold and (at t_1) the significance bar.

Partial Spearman ρ after controlling for brand age and competitive density: 0.41 at t_1, 0.43 at t_2. H4 falsified at the magnitude bar; t_1 also misses statistical significance.

H4 controls for two pre-registered covariates: brand age (years since product launch) and competitive density (market tier from the v0.6 registry, treated as ordinal). The control removes the most obvious confounds: older brands tend to have more consumer search interest and may have accumulated more AI training-data presence; incumbents differ structurally from challengers on both axes.

After this control, the H1 correlation falls from 0.50 to 0.41 (t_1) and from 0.48 to 0.43 (t_2). **H4 is falsified at the magnitude bar** (partial $\rho > 0.5$ required). t_1 also fails the significance bar (one-tailed $p = 0.066$); t_2 clears it ($p = 0.049$). The pattern points to real confound absorption: a meaningful component of H1's signal was age-and-tier covariation between AI Presence and search interest.

Per-row inspection confirms this. Workfront (24 years, mid-tier) has AI Presence 1.04% and Trends 6.67. Wrike (20 years, mid-tier) has AI 8.33 and Trends 12. Linear and Motion (both 7 years, challenger) over-perform their search-implied AI Presence; older mid-tier brands under-perform. The covariate control removes this systematic drift, exposing the partial relationship below it.

Limitations

Single category. v0.11 tests one of the five v0.6 baseline categories. The H1 just-miss, H3 falsification, and H4 covariate-controlled weakening are PM-software findings; their generalisation to running shoes, premium olive oil, premium facial skincare, and personal finance is the v0.12 hypothesis space, not a v0.11 claim.

Single validator. Google Trends search interest is one behavioural proxy for brand mindshare. Other validators — purchase data, brand-tracker survey measures, social-media share-of-voice — were not measured. A construct that doesn't correlate with one external proxy may correlate with another; v0.11's result is specifically about the AI-Presence-vs-search-interest relationship.

Small n. 17–18 brands per wave. The pre-registered floor was 16-of-19; v0.11 cleared it with margin, but the small-n constraint is real. The ρ-statistic's standard error is roughly 0.20 at $n = 17$, which is why H2's 0.02-magnitude stability is striking evidence of reproducibility rather than expected variance.

Trends index, not raw volume. Google Trends exposes only the bundle-normalised 0–100 search-interest index, not raw query counts. The pivot-rescaling protocol in v0.11 makes the indices cross-brand comparable, but the absolute query volume per brand cannot be recovered. The 'Asana = 100' baseline is structural, not substantive.

Two waves seven days apart. H2's stability claim is about the same construct measured twice across one week; it is not a claim about long-horizon stability. The v0.12 timing decision (next acquisition windows) will partly determine how far the H2 finding can be extrapolated.

What's Next

v0.12 — Three-category Phase 3 expansion. Add running shoes and premium olive oil to the PM-software protocol. The sharpened question: does the H1 just-miss + H3 leadership-zone divergence pattern generalise, or is it specific to PM-software's category-boundary structure? Premium olive oil's leadership zone is stable across both AI and consumer search (Brightland, Graza, Castillo de Canena) and is the cleanest replication candidate; running shoes will test the pattern under high-brand-density competitive conditions.

v0.13 — Full-category construct validity panel. Extend to the remaining two v0.6 baseline categories (premium facial skincare, personal finance). Personal finance carries the Mint phantom-brand effect documented in v0.7 and v0.10 and is expected to show the largest construct divergence.

Tri-System Brand Growth manuscript (Gonzalez Castro, 2026, in preparation). The v0.11 findings sharpen the framework's empirical content. The Linear paradox and Todoist inverse are candidate empirical anchors for the framework's distinctness claim — AI Availability as a third system distinct from Mental and Physical Availability.

Phase 4 (AIAS components 2–6). Construct validity established for AI Presence in v0.11–v0.13 is a precondition for measurement work on Ranking, Consistency, Coverage, Grounding, and Sentiment.

Hypothesis Scoring

All thresholds and tests locked at *v0.11-prereg* (commit f20ade8, 10 May 2026 UTC) prior to any Google Trends acquisition call against the wave windows. Pivot brand (Asana) exempted from at-acquisition E1b rule per pre-reg §5.1 (sd = 0 by pivot construction). Height (defunct September 2025) retained as phantom brand per registry-frozen-ness convention. n-floor evaluation: t_1 n = 17, t_2 n = 18 against the 16-of-19 floor.

H	Pre-registered prediction	Result	Status
H1	Spearman $\rho > 0.5$ AND $p_{1t} < 0.05$, both waves	$\rho_{t_1} = 0.496$ ($p = 0.021$); $\rho_{t_2} = 0.476$ ($p = 0.023$)	FALSIFIED
H2	$\lvert\Delta\rho\rvert$ 0.15	$\lvert\Delta\rho\rvert = 0.020$	**CONFIRMED**
H3	AI top-3 Trends top-5, both waves	1 / 3 at both waves	FALSIFIED
H4	Partial $\rho > 0.5$ AND $p_{1t} < 0.05$, both waves	partial $\rho_{t_1} = 0.41$ ($p = 0.066$); partial $\rho_{t_2} = 0.43$ ($p = 0.049$)	FALSIFIED
$H1_r$	Pearson $r > 0.5$ AND $p < 0.05$, both waves (sensitivity)	$r_{t_1} = 0.544$; $r_{t_2} = 0.533$	*CONFIRMED (sensitivity)*
$H1_{US}$	Spearman $\rho > 0.5$ AND $p < 0.05$, both waves (US sensitivity)	$\rho_{t_1} = 0.489$; $\rho_{t_2} = 0.455$	*FALSIFIED (sensitivity)*

Hypothesis Details

Per-hypothesis claim, operationalisation, and result. All confirm/falsify decisions use the pre-registered thresholds; the both-waves conjunction provides FWER control for H1 and H4 by construction (joint $p \approx 0.05^2 = 0.0025$ under the null).

H1 — Cross-Sectional Construct Validity. Per-brand AI Presence rate co-varies with per-brand Google Trends search-interest index at sufficient strength to support the construct. Spearman ρ across the eligible brand set per wave; confirmed if $\rho > 0.5$ AND $p_{1t} < 0.05$ at both waves. **Status: FALSIFIED.** $\rho_{t_1} = 0.496$ ($p = 0.021$); $\rho_{t_2} = 0.476$ ($p = 0.023$). Significance bar cleared both waves; magnitude bar missed at t_1 by 0.004 and at t_2 by 0.024. Direction supported; strength is not.

H2 — Correlation Stability. The H1 correlation strength is stable across $t_1 \to t_2$. $\lvert\rho_{t_2} - \rho_{t_1}\rvert$; confirmed if 0.15. **Status: CONFIRMED.** $\lvert\Delta\rho\rvert = 0.020$ — almost an order of magnitude below tolerance. The construct's measurement reproduces near-identically across waves. Whatever H1 captured at t_1 was captured again at t_2; the just-miss is not a sampling artefact.

H3 — Leaderboard Directional Consistency. Brands ranked top-3 by AI Presence are also among the top-5 by Trends, at each wave. Confirmed if all 3 of the AI-top-3 appear in the Trends top-5, at both waves. **Status: FALSIFIED at both waves.** 1/3 at each wave. t_1 overlap: Notion. t_2 overlap: Jira. Linear and Asana — the top two by AI Presence at both waves — fall outside the Trends top-5 at both waves. The construct divergence is concentrated at the leadership zone of the AI Presence distribution. See Finding 3 for the Linear paradox and Todoist inverse diagnostic.

H4 — Covariate-Controlled. The H1 correlation survives partialling out brand age and within-category competitive density. Pearson correlation on rank residuals after OLS on rank-transformed covariates (brand age years; tier ordinal incumbent=1 / mid=2 / challenger=3); df corrected for k = 2. Confirmed if partial $\rho > 0.5$ AND $p < 0.05$ at both waves. **Status: FALSIFIED.** Partial $\rho_{t_1} = 0.407$ ($p = 0.066$); partial $\rho_{t_2} = 0.428$ ($p = 0.049$). The H1 correlation loses meaningful strength under covariate control, indicating that age-and-tier covariation accounts for a non-trivial portion of the H1 signal.

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

Authored by

Pablo Ulpiano Gonzalez Castro

Principal Researcher, Third System

Faculty, MPS Branding Program, School of Visual Arts

Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.1). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

Gonzalez Castro, P. U. (2026). ***PM Software × Google Trends Construct Validity: AI Presence Index v0.11 — Phase 3 pilot***. Third System. thirdsystem.ai/v11-pm-trends-construct-validity

Underlying datasets

OSF project ec6wh, /v11/. Inputs (v0.9 AI Presence rates), Google Trends raw responses (10 pivot-bundle JSONs across 2 regions), topic-ID suggestion logs, pre-acquisition validation outputs, brand age source table, registry files, scoring outputs, build scripts, this report, and the matching SSRN working paper.

Cross-references: AI Availability foundational paper (SSRN 6659000); AIAS Presence Measurement Protocol v1.1 (SSRN 6722319); v0.6 Cross-Category Findings (SSRN 6720959); v0.7 Phantom-Brand Persistence Phase 2 BBB (SSRN 6721779); v0.8 Discourse-Language Knives (SSRN 6728000); v0.9 Longitudinal Re-Baseline (SSRN 6736878); v0.10 Naive-Phantom Rate Stability (SSRN 6741163).

Methodology log: v0.11 follows AIAS Presence Measurement Protocol v1.1 (unchanged from v0.9). Pre-registration locked at git tag v0.11-prereg (commit f20ade8) on 10 May 2026 UTC prior to acquisition. Acquisition session UTC timestamp captured in trends_acquisition_log.csv. No deviations from the pre-reg recorded as of publication..

Inquiries: hello@thirdsystem.ai

